MIL-T-13446D(MU)
20 August 1973
SUPERSEDING
MIL-T-13446C(MU)
20 May 1964

MILITARY SPECIFICATION

TELESCOPES, ELBOW: M16 SERIES, M116 SERIES, AND XM139

1. SCOPE

1.1 This specification covers elbow telescopes of the M16 series, M116 series, and XM139. These telescopes are similar except for differences in reticle patterns.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-P-116	Preservation, Packaging, Methods of
MIL-O-13830	Optical Components for Fire Control Instruments; General Specification Governing the Manufacture, Assembly, and Inspection of
MIL-F-13926	Fire Control Materiel; General Specification Governing the Manufacture and Inspection of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions

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DRAWINGS

U. S. Army Munitions Command

91-100-1	Telescope, Elbow: M16
A7597780	Telescope, Elbow: M16A1C
A7597781	Telescope, Elbow: M16A1D
A7597782	Telescope, Elbow: M16A1F
A7597783	Telescope, Elbow: M16A1G
A7597808	Telescope, Elbow: M16A1N
F8587431	Telescopes, Elbow: M116, M116D, M116F, M116C, and XM139

PACKAGING DATA SHEETS

PS 7597780	Packaging of Telescope, Elbow: M16A1C
PS 7597781	Packaging of Telescope, Elbow: M16A1D
PS 7597782	Packaging of Telescope, Elbow: M16A1F
PS 7597783	Packaging of Telescope, Elbow: M16A1G
PS 7597808	Packaging of Telescope, Elbow: M16A1N
PS 8289164	Packaging of Telescope, Elbow: M116D
PS 8289165	Packaging of Telescope, Elbow: M116F
PS 8587431	Packaging of Telescope, Elbow: M116
PS 8587491	Packaging of Telescope, Elbow: M116C
PS 10556120	Packaging of Telescope, Elbow: XM139

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Fabrication.- The telescope specified in the contract or order shall be manufactured in accordance with the applicable drawings listed in Section 2 and all drawings pertaining thereto.

3.2 General specifications.- The contractor shall be responsible for adherence to, and compliance with, the following requirements of Specifications MIL-O-13830 and MIL-F-13926.

3.2.1 MIL-O-13830.

(a) Cleanliness
(b) Illuminated reticle

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3.2.2 MIL-F-13926

(a) Order of precedence
(b) Dimensions and tolerances
(c) Inorganic protective surface finishes
(d) Part identification and Marking
(e) Workmanship

3.3 Environmental.

3.3.1 Storage temperatures.- The telescope shall show no evidence of physical failure and shall meet all performance requirements of this specification at a standard ambient temperature of 60° to 90°F after having been exposed to and thermally stabilized at ambient temperatures of -80° and +160°F.

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3.3.2 Vibration.- The telescope shall be vibrated at a constant frequency of 30 cycles per second with an amplitude of 1/16 inch (1/8 inch total excursion) for a period of 5 minutes ± 15 seconds. Subsequent to vibration the telescope shall show no evidence of loose, damaged, or missing parts, and it shall meet the requirements of 3.4 to 3.7 inclusive.

3.4 Cleanliness.

3.4.1 Dirt.- This requirement shall be met with the observations for dirt and foreign particles (classed as dirt) made through the eye end of the telescope. The maximum dimension of any particle appearing on the reticle surface shall not exceed the width of the reticle line. There shall be not more than three particles within the central 100 mil diameter field of view. There shall be not more than three particles outside this field of view. The angular separation between particles shall be greater than 15 mils. Particles smaller than 1/4 reticle line width shall be ignored regardless of distribution.

3.4.2 Other defects.- There shall be no evidence of moisture, grease, fingerprints, condensates, fractures, or adhesive separations when viewing through the objective end and the eye end of the telescope.

3.5 Sealing.- The telescope shall show no evidence of leakage when subjected to a pressure of 5 pounds per square inch gage pressure (psig) of dry nitrogen for a minimum period of 5 minutes. The dew point of the nitrogen shall be at least as low as -25°F.

3.6 Accuracy.

3.6.1 Collimation.- The M16 series, M116 series and XM139 telescopes shall meet the following requirements for the angle between the line of sight through the boresight cross and the geometric axis (established by the 1.375 inch diameter mounting surface):

		M16A1C	M16A1D	M16A1F	M16A1G	M16A1H
	AZ(Mils) EL(Mils)	0.0 0.0	0.0 0.0	0.0 20.4	0.0 33.4	0.0 18.2
M16A1K	M16A1N	M116	M116C	M116D	M116F	XM139
0.0 18.6	0.0 0.0	0.0 33.4	0.0 0.0	0.0 0.0	0.0 33.4	0.0 33.4

Tolerances on AZ & EL values shall be ± 0.5 mils. The face of the locating pad shall be in a vertical plane for all telescopes.

3.6.2 Eyepiece focus.- The reticle at the center of the field shall be in sharp focus when the eyepiece is set between -0.5 and -1.0 diopter.

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3.6.3 Reticle accuracy.- The angle between the boresight cross and the maximum range line shall be as indicated on the reticle drawing within 2.0 percent.

3.6.4 Parallelism of reticle and image.- With the axis of the 1.375 inch diameter mounting surface horizontal, as indicated on Drawing D7689539 for Telescopes of the M16 series and on Drawing F8587431 for telescopes of the M116 series and XM139, and with the face of the locating pad in a vertical plane, a horizontal reticle line 40 mils in length shall not deviate from a horizontal target line by more than the angle subtended by a 0.3 mil block at one extremity of the reticle line when the other extremity of the reticle line is in coincidence with the horizontal target line.

3.6.5 Image tilt.- With the telescope positioned as in 3.6.4, the image of a plumb line shall be vertical within one degree.

3.6.6 Resolution.- The resolution on the optical axis shall be 12 seconds of arc or less at one focus setting of an auxiliary telescope of at least 4 power positioned at the eyepiece. Resolution consists of detectable line structure and proper line count in all four meridians.

3.6.7 Parallax.- Parallax at the center of the field shall not exceed 0.1 mil when viewing a target at 250 meters ± 10 meters.

3.7 Illumination.- With the reticle illuminated, the reticle markings shall be discernible when observed in a darkened area.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article (initial production) approval.- The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of three Telescopes plus three each of all items covered by

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SQAP. The sample shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing.- When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials.

4.2.2 Contractor testing.- When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

4.3 Inspection provisions.

4.3.1 Submission of product.- Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.3.2 Examination and tests.- Components and subassemblies shall be inspected in accordance with the inspection provisions contained in Specification MIL-F-13926. Examinations and tests related to Section 3 herein shall be performed on a defect (individual characteristic) basis in accordance with Standard MIL-STD-105, and the inspection level and sampling plans specified in Table I titled "Classification of Defects". Examinations and tests for packaging, packing, and marking shall be in accordance with Specification MIL-P-116 and Section 5 herein. The tabulated classification of defects shall constitute the minimum inspection to be performed by the supplier prior to Government acceptance or rejection by lot. The Government reserves the right to inspect for any applicable requirement, and to reject individual non-conforming items.

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TABLE I. CLASSIFICATION OF DEFECTS

Use General Inspection Level II of Table I
With Sampling Plan Table II-A of Standard
MIL-STD-105

	REQUIREMENT	TEST PROCEDURE
Critical: None		
Major: AQL 1.0 percent defective		
101. Physical failure, post vibration	3.3.2	4.6.2
102. Cleanliness	3.4	4.7
103. Sealing	3.5	4.8
104. Collimation	3.6.1	4.9
105. Eyepiece focus	3.6.2	4.10
106. Reticle accuracy	3.6.3	4.11
107. Parallelism of reticle and image	3.6.4	4.12
108. Image tilt	3.6.5	4.13
109. Resolution	3.6.6	4.14
110. Parallax	3.6.7	4.15
111. Illumination	3.7	4.16
Minor: None		

4.3.3 Disposition of non-conforming product.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in Standard MIL-STD-105.

4.4 Control tests.- Three elbow telescopes shall be selected at random from each 50 produced, or from each month's production, whichever occurs first. Control samples shall have successfully met all acceptance tests specified herein prior to conducting the following examinations and tests:

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CONTROL TEST	REQUIREMENT	TEST PROCEDURE
Fabrication	3.1	Visual-Per applicable dwgs. and specs.
General specification requirements	3.2	Visual-MIL-F-13926 and MIL-O-13830
Storage temperature extremes	3.3.1	4.6.1

4.4.1 Control test failure.- Should the control test sample fail to meet the specified test requirements, acceptance of the product will be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted samples have been approved.

4.5 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment.- Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with applicable requirements specified in MIL-I-45607.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design.- All inspection equipment required by the contract and not provided by the Government shall be supplied by the contractor in accordance with the inspection equipment drawings forming part of the Technical data package.

4.5.2.2 Contractor design.- The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.6 Test methods and procedures.

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4.6.1 Storage temperatures.- The telescope shall be exposed to the temperatures specified in 3.3.1, utilizing test equipment in accordance with the "Test Facilities" requirements of Specification MIL-F-13926. This equipment shall be capable of maintaining the specified temperatures until thermal stabilization is reached. The temperature shall be changed gradually to avoid thermal shock. After the telescope has been thermally stabilized at each of the extreme temperatures, the telescope shall then be returned to the standard ambient temperature (60° to 90°F) and examined for evidence of damage or deterioration. The telescope shall then be subjected to the following test methods and procedures to determine compliance with the accuracy requirements referenced therein.

4.6.2 Vibration.- The telescope shall be mounted on a vibration machine equipped with facilities to securely hold the telescope while the machine is in motion. The telescope shall then be vibrated in a vertical plane at the amplitude, frequency, and time duration specified in 3.3.2. Subsequent to vibration the telescope shall be visually examined for broken, loose, damaged, or missing parts to determine compliance with 3.3.2.

4.7 Cleanliness.- The polished surfaces of the various optical elements in the telescope that can be distinguished shall be visually inspected for cleanliness in accordance with Specification MIL-O-13830 to determine compliance with the requirements specified in 3.2.1(a) and 3.4.

4.8 Sealing.- The telescope shall be placed in a pressure testing chamber similar to the chamber shown on Drawing F7680673. The chamber, with telescope, shall be attached to the Pressure Testing Fixture, Two Station - F8565556, or any approved equivalent external pressure testing medium. The chamber shall be pressurized with dry nitrogen gas to the required pressure and for the period of time specified in 3.5. Leaks in the sealing of the telescope shall be detected on the presure gage of the fixture to determine compliance with 3.5.

4.9 Collimation.- The telescope shall be mounted on a test fixture (fig. 1) and held on the 1.375 inch diameter bearing surface with sufficient spring-loaded clamping tension to allow the telescope to be rotated in three positions for collimating without slipping or rotating under its own weight. A device shall be on the fixture to insure that the housing or telescope tube is square when the locating lug on the tube and the device are in contact. With the telescope in this position, the bearing surface axis shall be horizontal within 10 seconds, and the vertical locating lug face shall be vertical within 10 seconds. A standard target collimator, F7573921, shall be mounted as illustrated (fig. 1). The collimator reticle shall permit reading or scanning of the entire field of the telescope. The horizontal and vertical lines of the telescope reticle shall be plumb within the

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tolerance specified when viewed in all positions. The geometric axis and the optical axis shall measure within the tolerance specified in 3.6.1.

4.10 Eyepiece focus.- The eyepiece of the telescope is a fixed focus type and shall be adjusted to a minus tolerance with respect to the reticle of the telescope. To determine the correct setting, a dioptometer of 3 power shall be used to test this requirement. The dioptometer shall first be adjusted to the user's eye by sighting through the dioptometer at a light or white background. Adjust the dioptometer's eyepiece until the reticle in the dioptometer is focused at the sharpest focus point obtainable. Rotate or slide the index on the barrel to the zero mark. Place the dioptometer against the eyepiece of the telescope. Sight through the dioptometer and the eyepiece of the telescope. Observe the reticle of the telescope and compare it with the reticle of the dioptometer. Rotate or slide the barrel of the dioptometer to make the reticle in the telescope go out of focus. Rotate or slide the barrel in the opposite direction to readjust or re-focus the telescope reticle to the same focal plane or sharpness as the dioptometer reticle. The reading of the dioptometer scale will indicate the error, which shall be within the tolerance specified in 3.6.2. The user should take several readings to insure conformance to the requirement.

4.11 Reticle accuracy.- The accuracy of the etched graduations on the reticle shall be in accordance with the reticle requirements and test procedures in Specification MIL-O-13830 to determine compliance with the requirements in 3.6.3.

4.12 Parallelism of reticle and image.- The telescope shall be mounted as in 4.9 and a line of sight through the telescope shall be directed to a target appropriate for measuring any tilt of the telescope reticle. Position one extremity of the reticle line coincident to a horizontal target line. The opposite extremity of the reticle line shall be coincident to the horizontal target line within the angular tolerance specified in 3.6.4.

4.13 Image tilt.- Image tilt in the telescope shall not exceed the tolerance specified in 3.6.5 as measured on a target appropriate for conducting this test. An auxiliary telescope, positioned with its vertical reticle line in coincidence with the vertical line of the target, is used in conjunction with the target for testing the telescope. The telescope shall be mounted as specified in 4.9. The auxiliary telescope shall be positioned at the eyepiece of the telescope. Sight through the auxiliary telescope and the telescope to the target. Image tilt will be recognized as tilting of the vertical target line image with respect to the vertical reticle line of the auxiliary telescope when the telescope is introduced into the line of sight.

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4.14 Resolution.- This test shall be performed in accordance with the resolution test in Specification MIL-O-13830, and the telescope shall meet the resolution value specified in 3.6.6.

4.15 Parallax.- Parallax shall be measured at the center of the field while observing a target positioned at the distance specified in 3.6.7. Any movement of the observer's head shall not cause the reticle lines in the center of the field to displace more than the tolerance specified in 3.6.7.

4.16 Illumination.- Utilizing illumination equipment providing a 3 volt ± .05 volt direct current source, reticle illumination shall be examined to determine compliance with the requirements specified in 3.7 when observed in a darkened area.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking.- Packaging, packing, and marking shall be in accordance with PS 7597780, PS 7597781, PS 7597782, PS 7597783, PS 7597808, PS 8289164, PS 8289165, PS 8587431, PS 8587491, or PS 10556120 as applicable. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use.- Elbow Telescopes of the M16 Series, M116 Series, and XM139 are utilized primarily as direct fire telescopes. All models are basically similar in design except for reticle pattern.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type of telescope being procured (see 2.1).
(c) Applicable stock number.
(d) Selection of applicable levels of preservation, packaging, packing, and marking.
(e) Applicable packaging data sheet (see 5.1).

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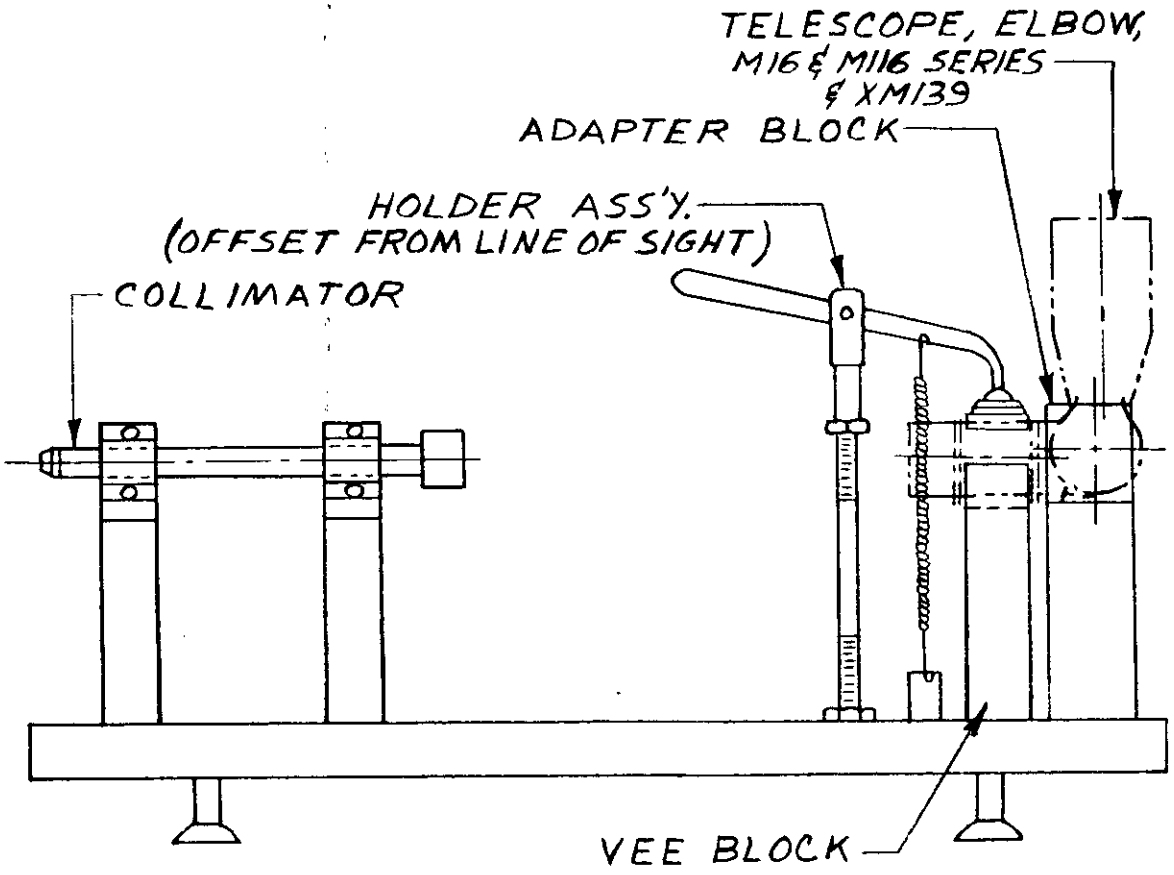
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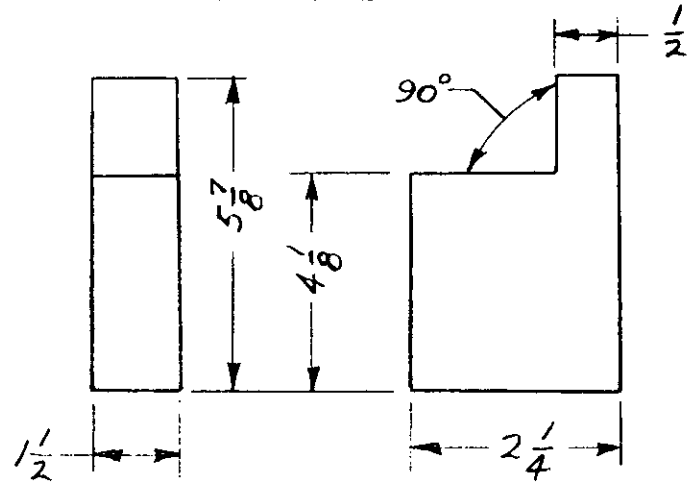


FIGURE 1. ADAPTER BLOCK FOR SUPPORTING ELBOW TELESCOPE.

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DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

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